Plant Station R18 - Design Review

Controlled field	Value
Document number	PLANT-TR-001
Plant design revision	R18
Document issue	D03
Issue date	11 October 2026
Preceding document	PLANT-TR-001 R17 D02
Status	Issued for combined prototype review; physical acceptance open

Current mechanical configuration: R18 open-route tower and R18 fascia with auxiliary holes closed; retained R16 lid (R15 geometry) and R13 planter. Electrical reference remains frozen Uno UI v6 and the 27 September interface record. Arduino/relay registration is Reported; physical retention, wiring, wet and operational acceptance remain open.

R18 combines the open pump-wire route and fascia aperture closures, retaining the R17 ports, Uno/relay mounts, lid, planter and Uno UI v6. Previous issues are retained; reported board fit is not acceptance of the new parts.

Revision control

Narrow parallel left-margin bars identify surviving changes against the verified matrix. Dates are documentary dates. R15/R16 source changes are incorporated without creating intervening manual issues.

Revision / class	Change and affected content
R5-R13 / source history	Earlier assembly and mechanical changes remain traceable through the preserved D01 matrix. Missing document editions are not invented.
R14 / preceding manual	AM R14 D01: Uno mounting and motor-disabled bench v2. TR's latest located same-family PDF is R13 D01; no TR R14 PDF is assumed.
R15 / 27 Sep 2026	Reinforced Uno bosses, thicker recessed-head lid and additional pump relief; source changes incorporated at R17, physical trials open.
R16 / 27 Sep 2026	Four relay bosses below Uno; source change incorporated at R17; actual underside and hole pitch unmeasured.
UI v6 / 27 Sep 2026	Three buttons and integrated lamps, enabled active-HIGH relay, 1-4 s request, soil display and 60 s backlight timeout. Firmware revision is independent of mechanical R17.
R17 / 28 Sep 2026	Mounted USB-C cut and pilots, adjacent diameter 10 DC extension hole; estimated dimensions authorised; coupon fit required.
D02 / 28 Sep 2026	Integrated current R17 document issue. Uses the established enhanced template, controls and cumulative revision bars. No R15/R16 manual issues created.
R18 / 11 Oct 2026	Combined open pump-wire route and closure of three standalone LED holes plus the speaker hole. Existing illuminated-button interfaces, board mounts and retained lid/planter remain. User reports Arduino and relay fit; independent screw, underside and route acceptance remain open.
D03 / 11 Oct 2026	Next controlled issue after R17 D02. Combined R18 files, partial reported fit evidence and new A18 acceptance records incorporated; A17 actions remain open. No new firmware, physical pass or accepted printer job is asserted.

D03 / document control: combined R18 technical content in the established A4 template. Pagination, new headings and editorial reflow alone receive no technical bar.

R5 is the earliest located assembly guide. R1-R4 and R9-R12 PDFs were not located in the previous inventory. R15/R16 are evidenced mechanical prototype changes, incorporated here without fictitious intermediate PDF/manual editions. Margin labels identify documentary changes, not physical test passes.

List of effective pages

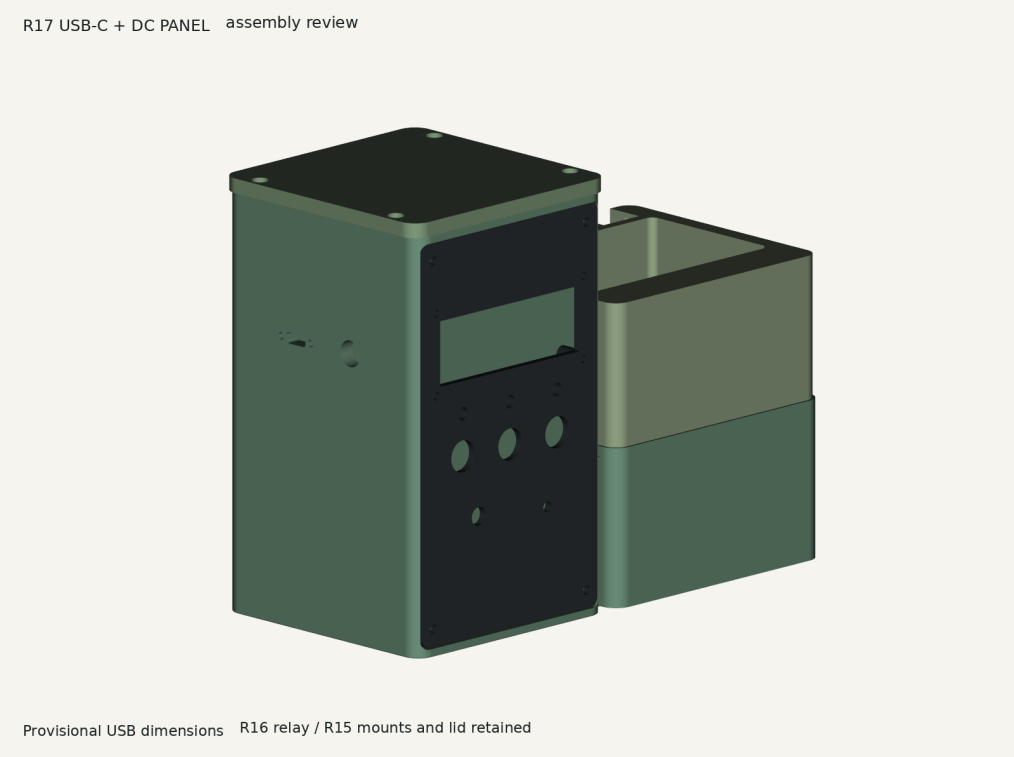
All pages are effective at D03, dated 11 October 2026, for the R18 prototype. Section references provide navigation; technical change labels are controlled separately.

Page	Issue	Content
1	D03	Control pages
2	D03	Control pages
3	D03	Control pages
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8	D03	Section 4
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Section / page cross-reference

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01 Current state and incorporated changes



Historical R17 assembled-STL review with the former enclosed route and R13 fascia apertures. It is retained for lineage and does not depict the combined R18 configuration or fitted hardware.

R18 R17 R18 combines the open pump-wire lay-in route and the fascia with three standalone LED holes plus the speaker hole closed. R15/R16 mounts/lid, R17 connector panel and frozen Uno UI v6 remain. Canonical tower/fascia exports are supplied for prototype review; physical and electrical acceptance remains open.

R18 R15 Historical R14 Uno supports reportedly split during screw insertion; R15 reinforced them. On 11 October Kevin reports the newest tower accepts both Arduino and relay perfectly but the terminated pump wire cannot pass the enclosed route. Preserve the fitted mounting geometry. No independent screw-retention, underside or powered-operation evidence closes the earlier actions.

R18 This R18 D03 issue succeeds R17 D02 and integrates the two previously separate candidates. R17 originals remain historical. New CAD validation is evidence for the digital geometry; the reported board fit is partial first-article evidence and no R18 physical print or fit pass is asserted.

02 Known constraints and configuration contract

Interface	Current basis	Evidence status
Body / fascia	R18 tower 207 x 100 x 158; fascia envelope 89.2 x 11 x 151.2	Canonical combined prototype; physical dry fit open
Lid	99 x 100 x 5.5; top z163.5	Confirmed retained R15/R16 CAD
Uno	Four asymmetric R14 centres; y76 seat	Source-verified; newest board fit Reported; actual underside open
Bosses	8.5 tip / 10 root / 19 projection; 2.8 x 11 pilot	Prototype; physical screw fit open
Relay	50 x 26 PCB; 44.5 x 20.5 pitch	Seller/photo nominal; underside unknown
USB panel	9.8 x 4.2 R1.2, 2.8 pilots at 15.2 pitch	Authorised best guesses
DC panel	Diameter 10; centre 28 from USB	User hole size; nut/grip unknown
Wet route	4 mm floor; tube bore retained; crossover floor z77, 17 above rim	Open lay-in channel geometry checked; no accepted fill or retention

Preserve the wet/dry partition, water bore, planter serviceability, asymmetric board registration, lid and retained control interfaces. User-edited print geometry was checked against the source before both candidates; no relevant source conflict was found. R18 changes no firmware or external supply architecture.

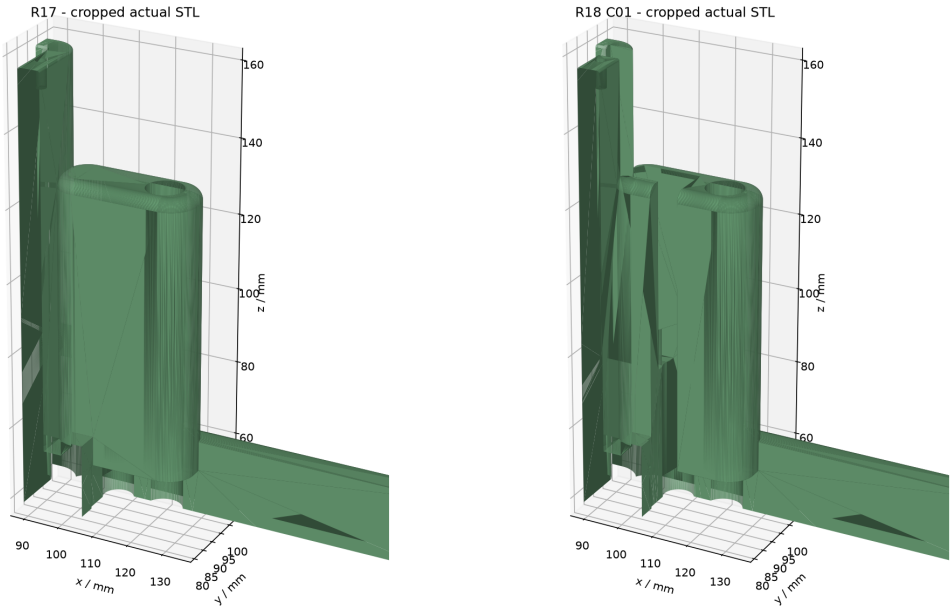
03 Mount, lid and pump options

Feature	Options and trade-off	Selected prototype / gate
Uno mount	Retain reported-fitting reinforced bosses; inserts/captive nuts would need new fit/access evidence	R15 mount retained in R18; reported board fit only, screw/underside trial open
Lid	Thicker pocketed lid gives flush heads; unchanged thin roof preserves old shape but misses request	R15 5.5 thick, 6.6 x 2.5 pockets; head/engagement trial
Pump-wire route	Enlarged closed bore depends on terminals; open lay-in avoids bore threading; added cover needs parts/fixings	R18 open front/top route selected within authorised scope; dry fit/retention open
Relay placement	Vertical below Uno consumes nominal height; side wall conflicts with access; rear-wall horizontal gives service space	R16 horizontal, 12 mm above saddle and 13 below Uno; terminal/wire fit open

Recommendation: retain the reported-fitting board mounts and combine the authorised lay-in route with the holes-only fascia change. Do not add a cover or alter button apertures. Validate actual wire bends, restraint, retained interfaces and print layers before accepting the prototype.

04 R18 route and fascia configuration decision

R18

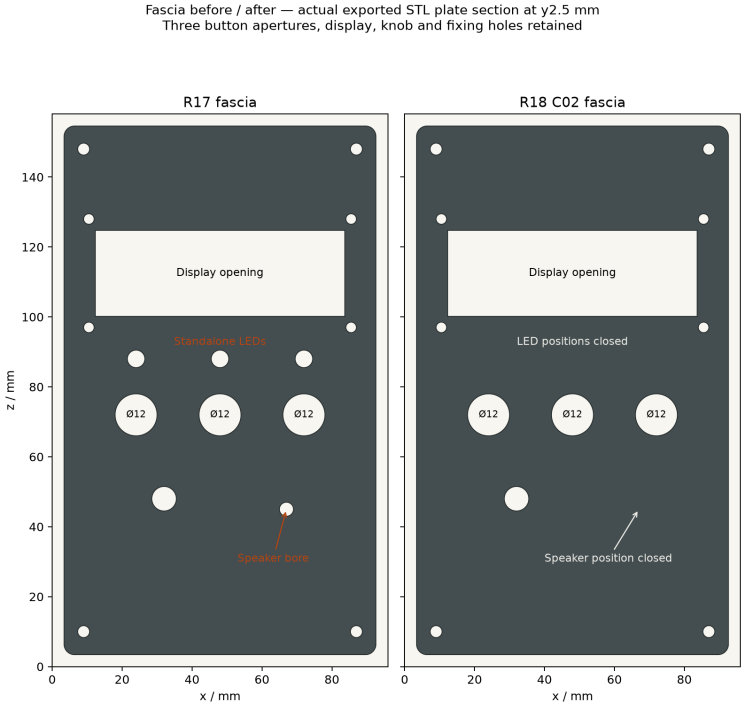


Actual R17/R18 mesh comparison of the enclosed and open pump-wire routes.

R18

Route cutters: front x106.5-114/y82-96/z52-129 and top crossover x89-114/y88-96/z77-159 mm. The crossing floor stays 17 mm above the 60 mm rim. Zero removed volume is recorded in the protected below-rim partition, sump floor, water-tube column, mounting region and rear-right lid fixing. Actual cable dimensions, bend radius, edge protection and retention remain Unknown.

R18



Actual fascia comparison: three 5 mm standalone LED apertures and one 4 mm circular speaker aperture closed; three illuminated-button openings preserved.

R18

Fascia option	Benefit / cost	Disposition
Fill only four obsolete apertures	Continuous 3 mm plate; retains fitted interfaces and rear geometry	Selected; replacement fascia, no new part

Fascia option	Benefit / cost	Disposition
Remove rear collars and piezo ring as well	Could simplify rear geometry; expands scope and requires new interface review	Outside holes-only instruction
Leave old apertures	Retains old part; leaves unwanted openings	Does not meet current request

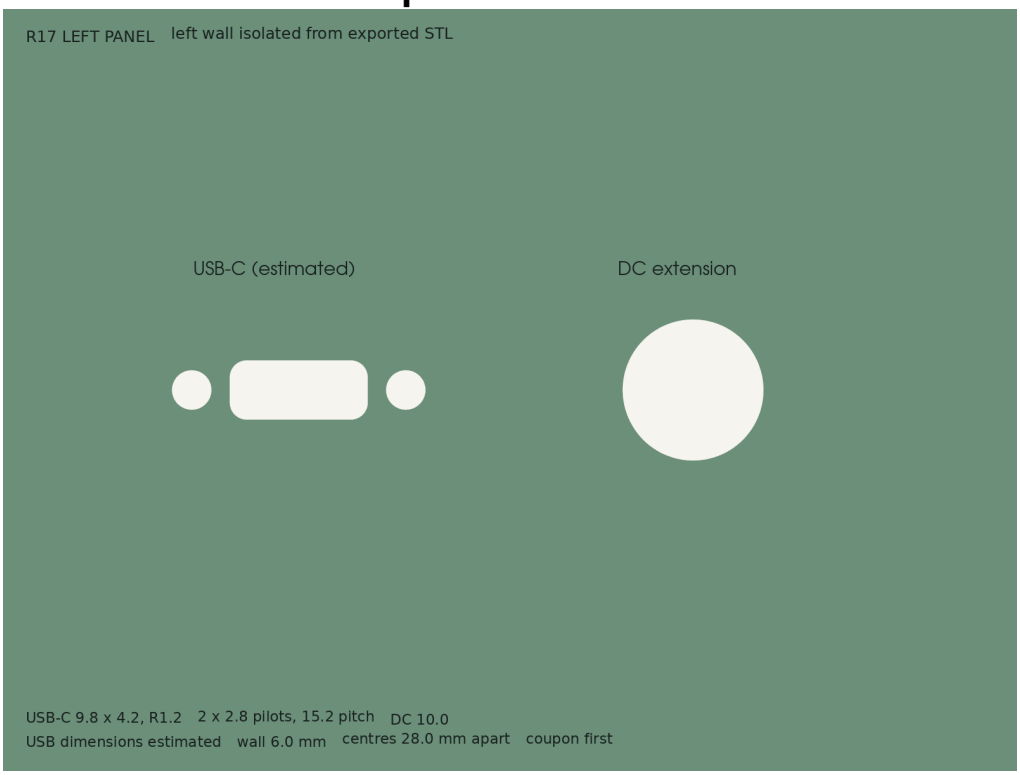
R18

R18

R18

Fascia fill is confined to y1-4 mm; 214.414 mm3 added and zero removed/outside the four patch regions. LED collars and piezo ring remain blind pockets. Three 12 mm button openings, LCD 71.4 x 24.6 mm aperture, 7 mm knob opening, four 3.4 mm fixings and display bosses remain. Integrated button LEDs retain D5/D6/D7 circuitry; D10 is unused.

05 USB-C and barrel panel decision



Left-wall mesh detail; both openings are at z108.5.

Option	Benefit	Trade-off
Integral wall cuts (selected)	Matches requested fixed panel; no extra printed part	Estimated USB fit and 6 mm DC grip must pass coupon
Replaceable connector insert	Easier later adjustment to another USB/DC component	Adds fastening, seams and another controlled part
Retain broad cable opening	Accepts direct original cable access	Does not implement requested fixed sockets

R17 The user requested the fixed USB/DC panel and explicitly authorised best guesses with PCB-sized holes. Integral wall cuts are selected for this prototype. USB shell cut 9.8 x 4.2 R1.2; pilots 2.8 through at 15.2 pitch; DC diameter 10. USB y63, DC y35, common z108.5.

R17 Estimated pilot-to-cut material is 1.3 mm. Image flange width 20.1 gives 12.95 mm to the DC cut; a guessed diameter-20 nut would leave 7.95 mm. These calculations depend on unmeasured component envelopes and do not establish fit or strength.

R17 **CAUTION**
Recommendation: coupon both actual connectors together, including mating plugs, nut grip, screw tips and solder service. Keep the electrical pin/power interface open until characterised; no USB-C rating or source behaviour follows from the cut.

06 Electrical and firmware configuration impact

R17 R14 disabled-motor bench v2 and spare third switch are historical. Current source enables active-HIGH D9 relay requests and uses D2/D3/D4 switches; D5/D6/D7 red/yellow/green lamps. A4/A5 serve LCD; A0/A1 serve probe/pot; D10 is unused.

R17 Dose requests are 1-4 s, knob sampled once. Yellow cycles pages/holds lamp test/cancels; STOP cancels and two-second hold clears session counters. Sensor is advisory; below 25% red blink, at/above green, raw <=20 fault. Software STOP and sensor fault do not provide hardware isolation.

R17 UI v6 commands soil Status return at request end and I2C backlight off after 60 s idle, waking on buttons or >=8 ADC pot-count change. Serial OFF/ON and LCD status 0 were recorded on 27 September; earlier communication failures and visual/post-dose checks remain open.

CAUTION
Relay module/pump rating, current, supply behaviour and reset default-OFF are unmeasured. USB-C pins/configuration and barrel polarity/grip are unresolved. CAD provision does not accept a new power system. Keep a physical pump disconnect.

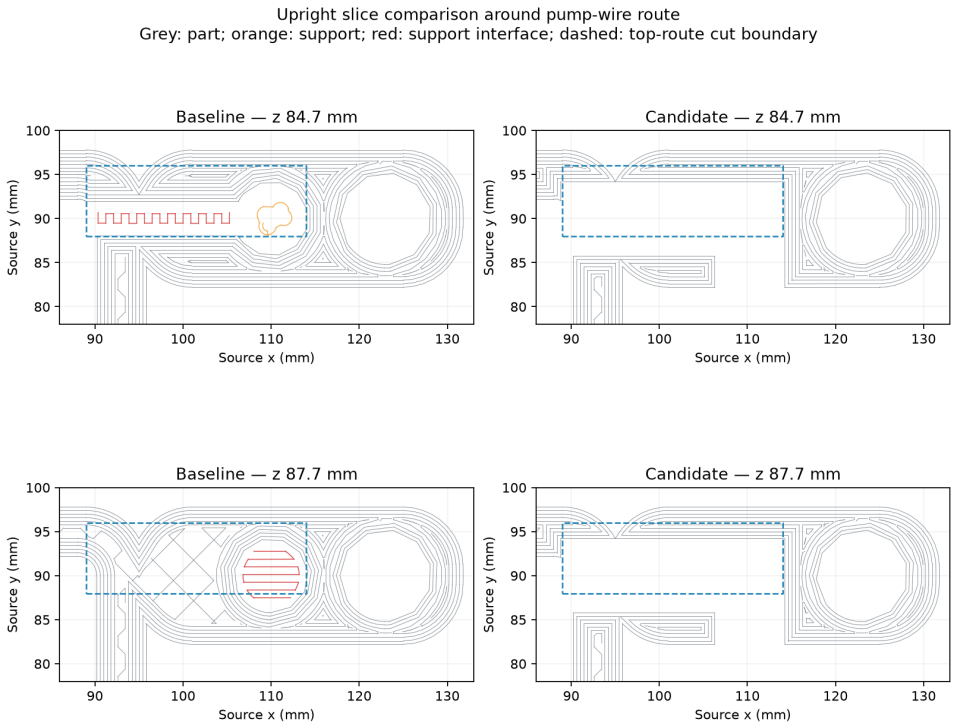
07 Evidence and digital verification limits

Evidence	Recorded result	Limit
R15 independent checker	Six final meshes/sections passed	Fastener, pump and leak physical gates open
R16 independent checker	Three single-component meshes and relay sections passed	Nominal board envelope only; underside unmeasured
R17 STEP/readback	Both exports reopened as one valid solid	Builder process still exits 1 on shutdown
R17 independent STL	Two closed consistently wound meshes; all edges two-face; no degenerate faces	Full mesh self-intersection not checked
R17 sections	Six wall sections: USB 9.800 x 4.200, DC approx 9.998 centre section, pilots approx 2.800	Tessellated mesh dimensions; not printed-hole readings
R17 change scope	0 added/removed mm3 outside left-wall mask	Does not establish physical insertion or strength
UI v6 source and record	Current SHA-256 matches 27 September upload record	No new hardware run in this documentation task

Combined nominal STEP geometry has zero intersection for all six main-part pairs and retains all eight board pilots. Actual solder, plugs, fasteners and service loops are not qualified. Original candidate CAD checks printed PASS then exited 1; scratch regeneration reproduced all three STL hashes, then both child processes returned 3221226356 (0xC0000374), wrapper exit 1. Combined STEP checks saved PASS then exited 1; mesh-only checking exits 0. A17-13 remains open; no full mesh self-intersection or physical acceptance is claimed.

R18 evidence	Recorded result	Limit
C01 route CAD/mesh	Valid single-solid STEP; closed wound single-component mesh; protected volumes unchanged	Actual cable, edge condition and retention unknown
C02 fascia CAD/sections	Valid single-solid STEP; 4 bores filled; retained sections/rear geometry coincide	New printed fascia dry fit open
11 October user report	Arduino and relay fit perfectly in newest tower	Reported registration; print hash/screw/underside evidence absent
Candidate slice comparison	Support estimate 24.215 to 23.699 g; approx 2.13% less in copied upright profile	Comparison only; preset/ZFiller/filament warnings and anomalous time metadata
Process and combined checks	Eight board pilots unchanged; 6 part-pair intersections zero; scratch STL hashes identical	Original CAD exit 1; regeneration child exit 0xC0000374 after PASS; wrapper/STEP checks exit 1, mesh-only exit 0

R18



Candidate comparison only: old enclosed route support/interface versus the opened route. These are not accepted R18 machine layers or printer instructions.

R18

The selected route tunnel ceilings are removed, but the full tower is not established support-free. Slicer support mass is estimated commanded deposition, not measured material use. Missing-system-preset, ZFiller and bed-temperature/filament warnings, plus implausible timing metadata, prevent a print-ready claim. R17 projects remain historical; the combined R18 set requires its own complete layer review.

08 Risks, unknowns and closure actions

ID / type	Condition and effect	Mitigation / closure
RI17-01 / Issue	USB cut/pitch and jack grip unmeasured; mismatch could require a body reprint	A17-01 coupon plus actual measurements
RI17-02 / Issue	R14 splitting history; newest Uno board fit reported, screw retention/strength still open	A17-02 representative screw/material evidence; inspect root/tip
RI17-03 / Risk	Relay solder/terminals and fascia services may collide	A17-05/06 full assembled dry fit and removal trial
RI17-04 / Issue	LCD intermittent communication faults; screen may fail under load	A17-09 supply/I2C/suppression investigation and repeat observations
RI17-05 / Risk	Relay closure through boot/reset or welded contacts may leave pump running	A17-08 measured default-OFF; physical disconnect
RI17-06 / Issue	Fill/leak/material limits unknown; ingress into electronics possible	A17-11/12 material qualification and controlled leak/drain-back/splash evidence
RI17-07 / Issue	USB/barrel wiring and rating unresolved	A17-07 identified source diagram, continuity and load evidence
RI17-08 / Issue	CAD exit code 1 prevents a clean-build claim	A17-13 isolate shutdown cause or retain explicit exception

R18

R18

Risk ratings remain qualitative; no rating matrix was supplied. All owners: Unassigned; target dates: Not set. A17 actions remain open. New risks are wire abrasion/lift at the open channel and defects or trapped services in the replacement fascia; close through A18-01 to A18-04 with actual build evidence.

09 Acceptance recommendation and next actions

R18 R17 1. Measure and lay in the actual terminated pump lead; check smooth bend, edge condition and retention. 2. Inspect the replacement fascia and dry-fit the existing buttons/display/knob/fixings. 3. Review complete R18 layers/material/orientation, then record the matching printed article and combined service/removal trial. 4. Carry forward all A17 connector, screw, electrical, default-OFF, LCD, leak/fill and dose gates before supervised wet acceptance.

R18 PLANT-TS-001 R18 D03 retains A17-01 through A17-13 and adds A18-01 through A18-04. Reported board fit does not close any complete action. Current disposition: combined prototype review; digital checks recorded; manufacturing, physical and electrical acceptance open.

R18 References: cumulative change matrix; preserved R17 D02 sources; R18 C01/C02 geometry, independent assurance, process logs and previews; reported_first_article.json; frozen Uno UI v6 and 27 September interface/upload evidence. No missing earlier manual issues are created.